DATA SCIENCE PROJECT · FUNDRAISING ANALYTICS

National FEC Donor Matching

Matching 552K university prospects to 6.5M political donors, with a confidence score for every match

~5,000

matched records added to a ~35,000-record prospect portfolio

Angelo Consorte · Data Science Intern · 2026

THE PROBLEM

The existing matches were regional and name-only

Before

One state only

Donors who moved or give from a second home were missed

First + last + city

No ZIP or street check, so common names collided

Undocumented rules

Conduit gifts (ActBlue/WinRed) possibly double-counted

After

National, 2022–2026

6.5M donor identities from FEC bulk files

Evidence-based score

Name, rarity, ZIP, street and household signals, 0–100

Documented and explainable

Every match shows the points behind its score

THE PIPELINE

From 150M+ donation rows to ~5,000 portfolio records

6.5M

FEC donor identities

Bulk files cleaned, conduit duplicates removed, names normalized

552K

prospects

Each with up to 4 addresses: contact, home, seasonal, business

~170K

candidate pairs

Blocked on last name + compatible first name, then scored 0–100

~5,000

portfolio records

New matches scoring 45+, plus earlier matches kept

Street addresses aren't in the bulk files, so they were pulled from the FEC API for candidate donors only.

THE SCORING ALGORITHM

How unlikely is this match to be a coincidence?

MAX POINTS PER SIGNAL (block width = weight)

35.3	29.4	23.5	+18	11.8
ZIP code	Name match	Name rarity	Street address	City / state

+8
Spouse bonus
Spouse also gives at the same ZIP

−2 / −3
Competing donors
Other plausible FEC people for the same prospect

−30
Address conflict
Street disagrees with every prospect address

≥ 85
Verified address
A confirmed same-address match is floored at Very High

CONFIDENCE LEVELS

Low <45 · not used	Medium 45+	High 60+	Very High 75+

HOW THE ALGORITHM EVOLVED

Four iterations, each fixing what the last one missed

v0 — v1 — v2 — v3

Reverse-engineer

Documented the old match: one state, first + last + city, no conduit filter.

National baseline

Name, rarity, ZIP density, spouse and collision logic. Caught two bugs that overstated giving by ~46%.

Street addresses

FEC API addresses, fuzzy-matched across all four address blocks: typos, abbreviations, PO boxes.

Edge cases

Maiden names, stray initials, hard-conflict caps. Refined with team and client review.

EXPLAINABLE BY DESIGN

Every match shows the points behind its score

20	9.4	35.3	11.8	+8	−3	=	81.5
Name	**Rarity**	**ZIP**	**City/state**	**Spouse**	**Competing**		**Baseline**
middle on one side	3 of 5	most recent	both agree	same ZIP	2 identities		

STREET CHECK

FEC 4120 MAPLE GROVE RD

Prospect 4120 Maple Grove Road

Same address after normalization: +18

99.5

Very High confidence

Illustrative example with fictional data

RESULTS

~5,000 records delivered and presented to the client

~5,000

records added to a ~35,000-record portfolio

93%

of new matches rated Very High

3,402

new national matches, plus 1,601 earlier matches kept



STACK & WORKFLOW

Python + DuckDB, accelerated by AI agents

Python + pandas

Normalization, scoring, explanations

DuckDB

Out-of-core processing of 150M+ rows

Colab + Cloud Storage

Notebooks and shared data

Claude Code

AI agents scaffolded each pipeline stage

What I'd keep: validate every AI-built step yourself.

Row counts at every stage, regression checks against the old matches, and a stratified manual review against FEC.gov.

All figures are aggregates. Client and donor data withheld.